

Corso di Laurea Magistrale in Ingegneria e Scienze Informatiche

Graph of Thoughts: A Runtime-Generated Graph for LLM Reasoning

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Relatore

Prof. Giovanni Ciatto

Candidato

Marco Raggini

Correlatori

Dott. Mattia Matteini

Prof. Andrea Omicini

Abstract

Large Language Model (LLM) have demonstrated exceptional capabilities in solving complex problems, yet they exhibit structural limitations that reduce their reliability such as opaque reasoning processes and limited exploration of alternative solution paths. Prompt engineering techniques, e.g. Chain of Thoughts (CoT), can improve reasoning process, however, they rely on a single linear reasoning path, limiting the scalability of output quality. This thesis introduces a Graph of Thoughts (GoT) architecture based on a runtime-generated graph, in which the LLM reasoning is guided by a directed graph constructed dynamically during execution. Such a graph contains various kinds of nodes, each representing a specific step: reasoning generation, tool invocation, response evaluation, and backtracking to alternative paths. Moreover, this approach supports the dynamic tool crafting mechanism when no available tool is sufficient to solve the problem. A dedicated agent generates and registers new tools at runtime, extending the system’s capabilities without requiring human intervention. Intermediate reasoning process is evaluated by an LLM judge with an adaptive threshold based on the estimated complexity of the problem. The framework is evaluated on the Hendrycks MATH benchmark, comparing the proposed architecture against a Zero-Shot CoT baseline across three different Gemini models. The results suggest that the GoT can make more robust models with stronger reasoning capabilities, where the backtracking mechanism and multi-path exploration help recover from incorrect intermediate steps.

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Chapter 1

Introduction

Write your intro here.

You can use acronyms that you defined previously, such as Internet of Things (IoT). If you use acronyms twice, they will be written in full only once (indeed, you can mention the IoT now without it being fully explained). In some cases, you may need a plural form of the acronym. For instance, that you are discussing Virtual Machines (VMs), you may need both VM and VMs.

Marco Raggini: Add sidenotes in this way. They are named after the author of the thesis

Structure of the Thesis

Marco Raggini: At the end, describe the structure of the paper

Chapter 2

Background

2.1 Large Language Models

LLMs are a class of models which belong to Deep Learning, a sub-category of the broad field of Artificial Intelligence (AI). In recent years, they have become extremely popular thanks to their ability to understand and generate natural language with a high level of fluency and coherence [ZZL⁺26]. Their generalized knowledge and versatility make them perfect for a wide range of applications, from text generation and translation to code assistance, question answering, and conversational systems.

The ability to generate natural language fits perfectly with the need to create systems that can be used by everyone without requiring specific technical knowledge. LLMs allow users to interact using natural language, which enables to access them through simple conversational interfaces, making human-computer interaction more intuitive and accessible.

During the last years, after the release of ChatGPT, the average number of arXiv papers that contain the “*large language model*” title or abstract are deduplicated [ZZL⁺26], leading to the development of increasingly powerful models such as

GPT-4 [Ope23], LLaMA [TMS⁺23], Gemini [Tea25], and Claude ¹. These systems are capable not only of understanding and generating text, but also of solving reasoning tasks, summarizing documents, writing code, and supporting decision-making processes, becoming more and more used in real-world scenarios.

2.1.1 How they work

LLMs are autoregressive models that simply try to predict the next word in a sequence of characters, based on the context.

The text given as input to the LLM is preprocessed and divided into smaller units called *tokens*. A token may represent a word, part of a word, or even a symbol, depending on the strategy chosen. The next step is to convert the tokens into numbers, in such a way that a neural network can process them.

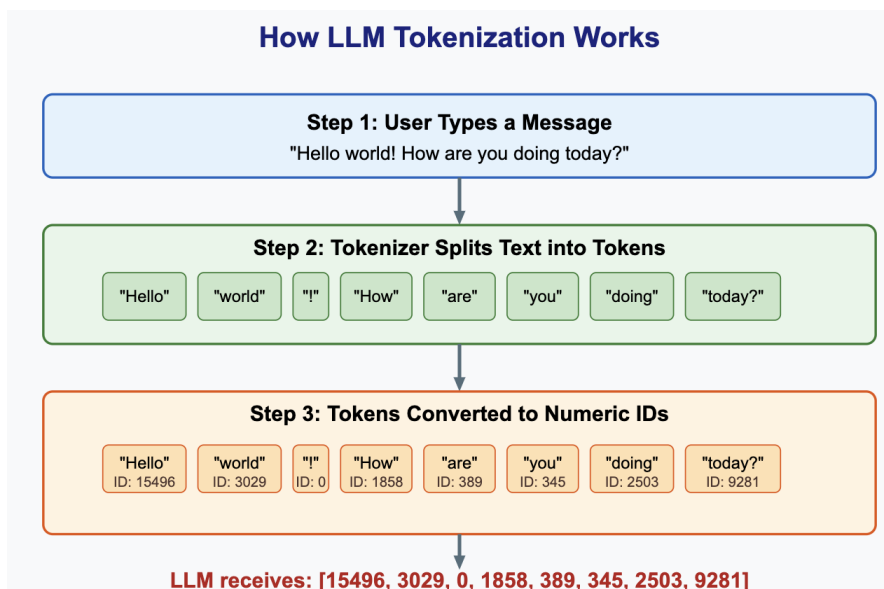


Figure 2.1: Schematic workflow of an LLM tokenizer: from raw input text to token segmentation and subsequent mapping into vocabulary IDs. Source: <https://www.linkedin.com/pulse/llm-tokenizers-hidden-engine-behind-ai-language-models-natarajan-rmbsc>

Modern LLMs are typically based on the Transformer architecture introduced

¹<https://assets.anthropic.com/m/61e7d27f8c8f5919/original/Claude-3-Model-Card.pdf>

by Vaswani et al. [VSP⁺17]. The Transformer uses the self-attention mechanism to capture contextual relationships between tokens, allowing the model to understand long-range dependencies and generate coherent text.

Once the tokenization is done, the text generation starts using autoregressive algorithm: a token is generated and appended to the input sequence and used to predict the following token. By repeating this process multiple times, the model can generate complete sentences, paragraphs, or entire documents.

Despite the apparent intelligence of these systems, LLMs do not possess real understanding or reasoning capabilities in the human sense. Their outputs are generated through statistical pattern matching learned from training data, which explains their tendency to occasionally produce incorrect or hallucinated information [XJK25] and the standardized phrases that often generates.

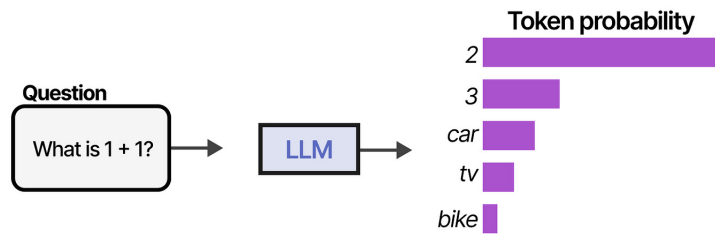


Figure 2.2: Schematic overview of autoregressive decoding and probabilistic next-token prediction in an LLM. Source: https://albanna-tutorials.com/llm_agents.html

2.1.2 Prompt Engineering

Prompt engineering is a relatively new field of research that refers to the practice of designing and implementing prompts or instructions trying to guide the LLM output. The main goal of prompt engineering is to optimize the LLM output by solely refining the prompt [Mes23].

The prompt could include a lot of components, such as: *Directives*, *Examples*, *Output formatting*, *Style instructions*, *Role* or *Additional Information* [SIB⁺24]. All of these components can help to improve the generated output in terms of a better style, accuracy in the response, or even a more correct output.

One of the most popular and older prompt engineering techniques is the *Few Shot Prompting* [BMR⁺20], in particular, it has three sub categories:

- **Few-Shot:** We are referring to a prompt that includes a few examples of how the output should be structured with also the correct output. This context helps the LLM to learn how to complete the task given without updating the model parameters.
- **One-Shot:** It is equal to the few-shot prompting, with only one demonstration provided and the natural language description of the problem.
- **Zero-Shot:** In this case, no demonstrations are provided. The prompt contains only the natural language description of the task. This method has the maximum convenience, but could be challenging to apply, mainly due to the difficulty of imagining the format expected without producing examples.

These methods are not meant to compete with each other. There are three possibilities depending on the task we are facing. In general, few-shot prompting tends to perform better on structured or domain-specific tasks, while zero-shot prompting is often sufficient for simpler or well-known tasks [BMR⁺20].

2.1.3 LLM-as-a-judge

Mattia: rifaserei meglio questa sezione, prima partirei dal bisogno di un LLM-as-a-judge e perchè è stato ideato, poi spiegherei cosa è, infine pro e contro

During the prompt engineering discovery, a new way of improving the output is created, the LLM-as-a-judge technique [ZCS⁺23]. The idea is to use Generative AI to judge another LLM output; this method can benefit the output in scalability, explainability and has the advantage of limit the human involvement of the reasoning process.

The three approaches proposed to use the LLM-as-a-judge are:

- **Pairwise comparison** The judge is presented with one question and two or multiple answers. Its goal is to choose which one is better.

- **Single answer grading** It is directly asked to the LLM judge to assign a grade to the solution.
- **Reference-guided grading** In some cases, providing a reference solution may be beneficial.

This technique has also several bias such as **Position bias**, **Verbosity bias** and **Self-enhancement bias**.

2.1.4 Agents

The LLMs are now good enough to start thinking about how to automate some processes or how they can be used in more contexts. For these reasons, the agents are created. The formal definition is:

“An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through effectors.” [RN10]

A high-level description of an agent consist of four main components:

- **Senses** the surrounding environment around it.
- **Perceives** the information of that environment.
- **Reasons** about how to act based on the input given.
- **Acts** upon the environment through its actions (the output) [RN10].

In the LLM context, the agent brain is supported by the LLM, which reasons using the context given (it could be the environment as a text), and it acts generating an output or by calling the **tools**.

Tools are external functions or Application Programming Interfaces (APIs) that an agent can invoke at runtime to overcome the limitations of the LLM, such as the lack of up-to-date knowledge, the inability to execute code, or the absence of persistent memory. By selecting and calling the appropriate tool based on the user’s intent, the agent extends its capabilities well beyond pure language generation.

As previously mentioned, LLMs can be unpredictable and lacks of transparency. In this sense, the usage of tools enhances the credibility of the decision of an agent, improving interpretability and robustness [XCG⁺23].

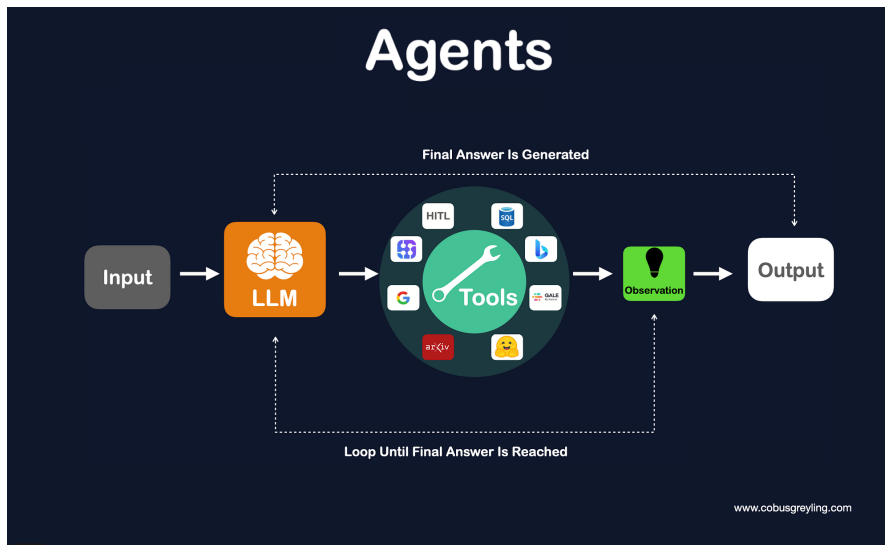


Figure 2.3: Representation of how an agents works. Source: <https://cobusgreyling.medium.com/how-would-the-architecture-for-an-llm-agent-platform-look-b07d7e004561>

A difference between pure LLMs and AI agents is the autonomy. While LLMs goal is to generate a correct and coherent output, the agents try to perceive the autonomy, minimizing the need for human intervention, and trying to adapt by changing their behavior based on feedbacks [Kum24].

Mattia: se metti una figura la devi citare da qualche parte

2.2 LangChain & LangGraph

2.2.1 LangChain

LangChain is an open-source framework launched in 2022 and designed to simplify the development of applications powered by LLMs. The LangChain framework has two main declared core focuses ²:

API Abstraction Different providers expose different APIs. The first goal of the framework is to standardize the way to interact with the main LLMs providers. This approach allows developers to focus on the application logic rather than dealing with the low-level details of each provider.

²<https://docs.langchain.com/oss/python/langchain/philosophy>

Tool Integration LLMs are more than text generators, and the research is moving to the direction of creating agents that interact with the environment and communicate with tools. The second goal is to provide a simple way to define and integrate tools that LLMs can use, it includes also the possibility of parsing and access to unstructured data.

The framework is designed with modularity as a core principle. Rather than being distributed as a single package, LangChain is split into independent components that can be installed separately, such as `langchain-core`, `langchain-community`, and provider-specific packages like `langchain-google-genai`. This allows developers to include only the dependencies strictly required by their application.

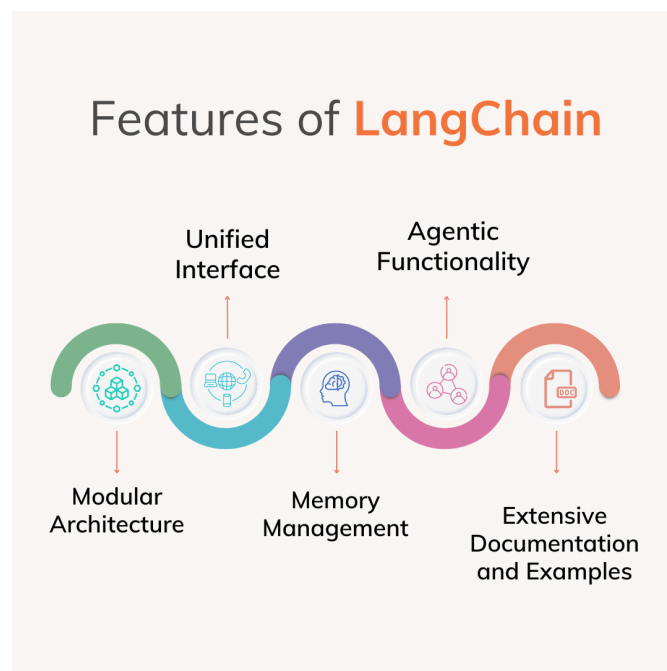


Figure 2.4: Representation of the LangChain features. Source: <https://www.ksolves.com/blog/artificial-intelligence/power-of-langchain-features-and-benefits>

2.2.2 LangGraph

In 2024, a new module called LangGraph was released ³. Rather than LangChain, which focuses on simplifying the interaction with LLMs and neglects low-level details, LangGraph is a low-level framework, focuses entirely on agent orchestration. With LangGraph, developers can define complex agent workflows. The core abstraction of LangGraph is the *stateful directed graph*, where each node represents a computational step and edges define the transitions between them. Transitions can be either deterministic, where the next node is uniquely determined by the completion of the current step, or conditional, where the next node is selected dynamically based on the output of the current one. This makes LangGraph particularly well-suited for systems that require non-linear control flow, such as backtracking, branching, or iterative refinement of intermediate results.

2.3 Related Works

While few-shot prompting improves performance by providing input-output examples, it does not explicitly guide the model through the reasoning process required to solve a task. The research in this field has been evolved, finding new ways to improve the output quality without changing the LLM parameters.

2.3.1 Chain of Thoughts

Chain of Thoughts is an important prompting technique introduced in [WWS⁺22]. CoT reveals that, when LLMs produce the reasoning steps of a solution, the correct results increase dramatically. This leads to several advantages:

- **Accuracy:** As we said previously, the number of corrected answers increase [WWS⁺22].
- **Interpretability:** The LLM remains a black box, but the final solution includes the steps, allowing to understand when the LLM produces an error in the reasoning.

³<https://docs.langchain.com/oss/python/langchain/philosophy>

- **Transparency:** In some task categories, such as math problems, the CoT techniques produces the intermediate reasoning which helps to independently reproduce the result given.

The CoT was firstly introduced as a few-shot approach, where the examples were used for inducing the multi steps reasoning. Recently it has been proved that also a zero-shot CoT increases the performance, introducing the “Let’s think step by step” phrase into the prompt [KGR⁺22].

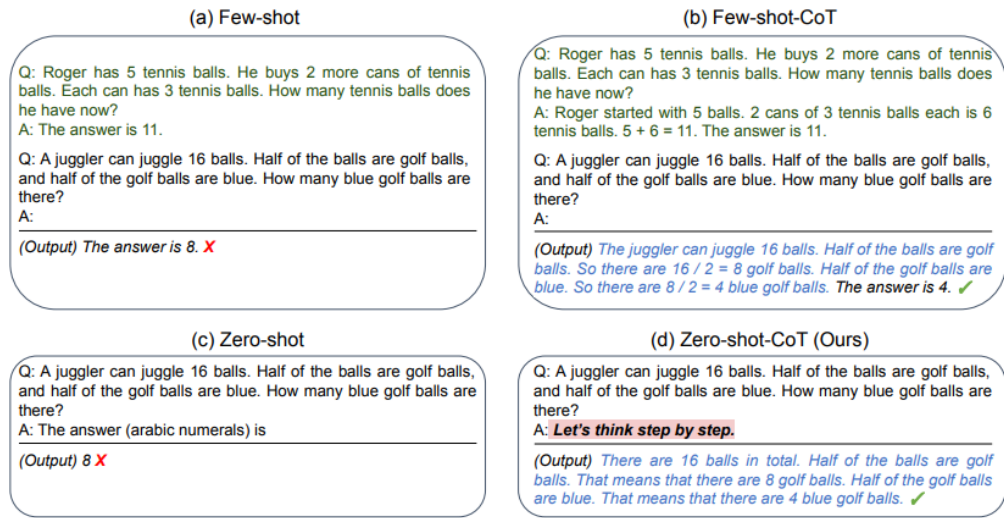


Figure 2.5: The differences between a normal prompt versus a few-shot CoT and a zero-shot CoT. Source: [KGR⁺22]<https://arxiv.org/abs/2205.11916>

Later on, the Self consistency CoT (CoT-SC) [WWS⁺23] was developed as an evolution of the classic CoT. The idea behind that was explore multiple reasoning paths in order to select the most accurate output. The empirical evaluation proved that CoT-SC boost the performance of CoT reasoning [WWS⁺23].

2.3.2 Tree of Thoughts

The introduction of the CoT approach paved the way for a new type of prompt engineering. From this method, the Tree of Thoughts (ToT) was developed [YYZ⁺23]. While the CoT produces multiple steps with a single prompt, the ToT is based on the idea of generating multiple thoughts for each step and evaluating the best

of them; each step is a partial solution and the evaluation phase guarantees the refinement of it into the final response. In order to achieve the best result, the architecture includes four important parts:

1. **Task decomposition:** In the ToT, the task decomposition is an explicit part. This should helps the LLM to decompose the problem until they are small enough to generate a satisfying solution.
2. **Thought generation:** There are two ways of generating thoughts. The *Samples* is useful when the solution space is rich and the *Propose* is more efficient when the solution space is restricted.
3. **State evaluator:** Two strategies were proposed in order to evaluate the thoughts. In the *Value* strategy, the LLMs give a value to each thought from 1 to 10 and chose the higher one. In the *Vote* strategy, each evaluator choose the best thought and who takes more votes become the winner.
4. **Search algorithm:** It is used to search the more accurate solution in the tree; the strategies proposed are the classics, Breadth-first search (BFS) that controls each node at every level of the tree, Depth-first search (DFS) that explores the most promising state first until the final output is reached.

Mattia: parli di Samples e Propose ma non dici cosa sono

The studies proved that this approach obtains better results than CoT, CoT-SC and the not refined prompt [YYZ⁺23].

2.3.3 Graph of Thoughts

Another evolution of the CoT is the GoT architecture [BBK⁺24]. The authors propose an architecture where the LLM reasoning process is seen as a graph of thoughts, where each vertex is represented by a thought and the edges are the dependencies. This structure allows the creation of different operations: **Generation**, which produces new thoughts from one or more existing ones; **Aggregation**, which merges multiple thoughts into a single one; and **Refining**, which iteratively improves an existing thought based on self-feedback.

Technically, the reasoning process consists of two main parts:

Graph of Operations (GoO). It is the planning part of the system. The GoO is constructed before the execution starts and it can not be changed during the reasoning process. As the name suggests, it contains all the operations which will be completed during the execution. Importantly, the GoO instance is not generated by the LLM, but is created by the user at the very beginning.

Graph Reasoning State (GRS). It represents the graph that maintains all the information updated during the execution. The LLM follows the instruction of the GoO, the output produced is stored in the GRS. Specifically, the GRS tracks the executed operations, the LLM outputs, the score and validity of each thought, along with additional metadata generated throughout the reasoning process.

Although the architecture just explained cannot be easily deployed in real-world scenarios by non-expert users due to the requirement of manual GoO creation, research demonstrates that a proper implementation of GoT can significantly outperform previously discussed architectures such as CoT and ToT [BBK⁺24].

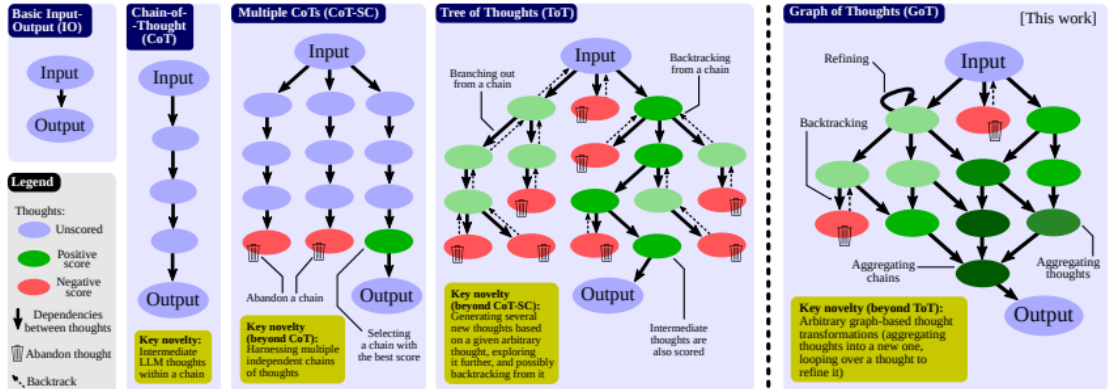


Figure 2.6: Representation of different prompt engineering architectures. Source: [BBK⁺24]<https://arxiv.org/abs/2308.09687>

Chapter 3

Design

This chapter presents the proposed approach in detail. We first formally define the problem and highlight the main challenges and requirements. Then, we describe the proposed GoT, providing its formalization, including algorithmic details and execution strategies. Lastly, we illustrate the approach with a concrete example to demonstrate its application.

3.1 Problem statement

3.1.1 Task Definition

Before introducing the proposed approach, we first formally define the problem addressed in this work.

Given an input problem P expressed in natural language, the objective is to generate a solution S along with a structured reasoning process R that leads to S . The reasoning representation R is modeled as a GoT framework that is automatically generated by the architecture, bypassing the need for manual design (as the GoO described in section 2.3.3). The overall reasoning process should be coherent, interpretable, and effective in reaching correct answers.

3.1.2 Challenges and Requirements

LLMs are able to generate coherent text and perform reasoning, but they carry a set of limitations that can be mitigated. In this work, the main challenges that we want to address are the following:

- **Limited explorations of alternatives:** The standard approaches, such as CoT and ToT generate a single linear solution, which may not be sufficient to generate a good solution, especially for complex problems that may require more attempts.
- **Black box reasoning:** The reasoning process of LLMs is often difficult to understand and interpret. Since generated text is autoregressive and the model does not maintain an explicit representation of its reasoning process, debugging and improving the model becomes difficult.
- **Error propagation:** If there is an error in the early steps of the reasoning process, it will be propagated to the final solution. Without more exploration of alternatives, the model may not be able to recover from these errors and may produce incorrect solutions.

In order to address these challenges, the requirements are formalized as follows:

- **Exploration of alternatives:** If the LLM answer does not satisfy evaluation criterias, the model should be able to explore alternatives, which may require the ability to backtrack and try different reasoning paths.
- **Separation of reasoning:** The system should allow starting a new reasoning process independently if a path leads to an incorrect solution, without affecting other trajectories.
- **Structured and interpretable reasoning:** The reasoning process should be captured in a structured representation that enables traceability, interpretability, and evaluation of intermediate steps.

3.1.3 Assumptions

In order to simplify the problem and focus on the core aspects of the GoT paradigm, we make the following assumptions:

- **LLM capabilities:** We assume that the LLM has enough capabilities to use external tools to craft tools if needed to evaluate the generated solutions and to generate intermediate thoughts that are coherent and relevant to the problem at hand. This assumption is strictly dependent on the size of the LLM that we are using.
- **Tool-Centric Problem Domain:** We assume the input problems belong to a class of tasks that either cannot be solved through direct, zero-shot text generation alone, or can significantly benefit from external tools in terms of reasoning accuracy and output interpretability. Consequently, these problems are assumed to require a reasoning sequences and at least one interaction with an external environment or tool (e.g., a calculator, a search engine, or a code interpreter).
- **Resource Finiteness:** We assume the existence of an upper bound on the graph depth and the number of LLM invocations to ensure the termination of the process and to mitigate potential infinite loops caused by recursive backtracking. These assumptions is also valid for time and computational resources, which are limited and should be taken into account when designing the system.

3.2 The proposed approach

The approach proposed in this thesis is based on the GoT paradigm [BBK⁺24]. In this framework, the reasoning process is modeled as a graph, where nodes represent intermediate thoughts, and the LLM guides both the expansion of the graph and the decision of when to terminate the reasoning process.

Given an input problem P , the system incrementally constructs a graph according to a set of high-level rules. These rules are designed to balance the exploration

Mattia: non penso possiamo assumere che il grafo generato abbia una profondità limitata, e anche se lo facessimo, si risolverebbe con un meccanismo di early termination o fallback. l'assunzione ha senso invece sulle risorse computazionali e sul tempo.

of multiple reasoning paths with the need for coherent and effective problem-solving. Additionally, they encourage the use of external tools, which play a key role in improving traceability and interpretability of the reasoning process.

3.2.1 Motivations for using GoT

The GoT paradigm is chosen for the potential that this structure offers in terms of exploration, combination of different reasoning paths and the ability to backtrack in order to recover from errors.

The second motivation about the use of GoT, as mentioned in the section 2.3.3, is the fact that the GoT research is still in its early stages, so its potential is not yet fully explored.

Mattia: potresti allungare questa sezione parlando di exploration, pro sull'uso dei tool, backtrack, tool crafting, possibile "cache" dei tool creati, ecc.

3.2.2 Static Graph Structure

As a **static graph**, we refer to the graph structure that is defined at the beginning of the process, which includes the definition of nodes and edges, as well as the rules for how they can be connected. Figure 3.1 illustrates the static graph structure, which provides a framework for how the reasoning process can be represented and how different thoughts can be connected to each other.

Mattia: controlla tutte le altre immagini e assicurati di citarle nel posto giusto, e dove puoi ritaglia i contorni (togli lo spazio bianco) per guadagnare spazio

The static graph structure is defined as $G_s = (S, T)$, where S is the set of possible states (e.g. Evaluation, Backtracking, Crafting etc.) and T is the set of possible transitions between these states. The states are the following:

__start__ and __end__. The **__start__** is the first state reached and the **__end__** is the last reached in the process. They are respectively the entry and exit points and are visited exactly once per execution.

goal. This is the first state immediately after the entry point. The system starts with the user prompt and the **goal** represents the problem. From here, the system starts to generate thought paths that can be explored in order to produce a solution.

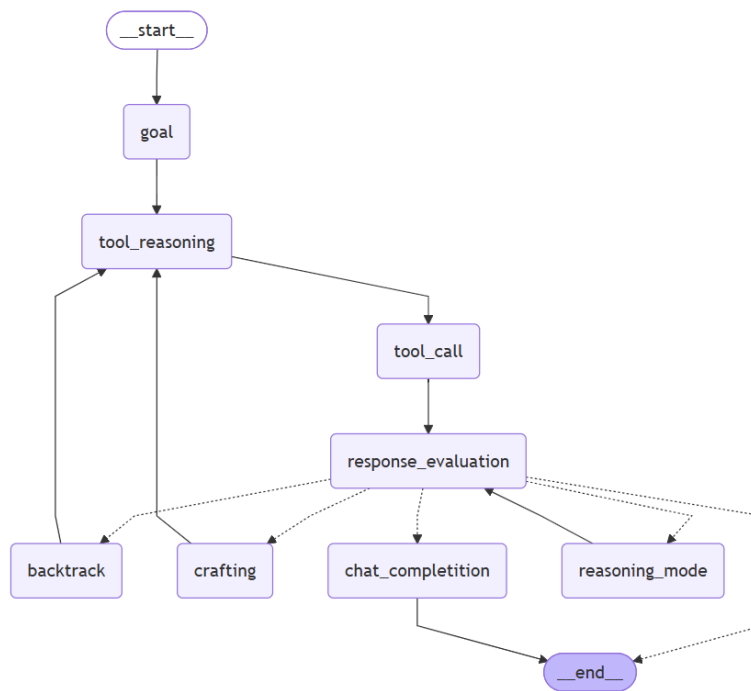


Figure 3.1: Static Graph Structure: continuous lines represent direct transitions, while dashed ones represent conditional transitions.

In particular, the system creates four different reasoning paths composed of three `tool_reasoning` nodes and one `reasoning_mode` node. The number of tool reasoning paths created are arbitrary; they are chosen by observing the system's behaviour during its runs. A higher number of reasoning paths can lead to more solutions, but the cost in terms of time and tokens increases consequently.

Mattia: se il numero di paths creati è arbitrario, perchè prima dici che son 4? poi non si capisce dove vengono creati sti paths... intendi nel dynamic graph? se così non va bene perchè non è ancora stato introdotto.

tool_reasoning. This is the starting point of a reasoning path. From here, the system starts to think about how to address the problem without solving it directly in order to force the LLM to reason about the it. The main focus of this state is about how to use the tools or, if necessary, how to craft them. The message history will be used to provide the context in the following states.

Mattia: questa frase non è vera per tutti gli stati?

tool_calling. This state is reached after the `tool_reasoning` state and it represents the moment in which the system should call the tool or just generate the solution if the tool is not needed. Since LLMs are intrinsically non-deterministic, it is not possible to predict the exact output; however, the system is designed to guide the LLM towards the use of tools through the provided context. The expected behavior is for the model to generate a response that incorporates tool calls (if deemed necessary) or a direct solution, effectively bridging the gap between reasoning and action.

response_evaluation. It is reached after `tool_calling` or `reasoning_mode` states, and it represents the moment in which the system evaluates the generated response of the reasoning process. This step is crucial because the evaluation determines whether the response is ready to be accepted as a solution, whether the system needs to backtrack and explore alternative reasoning paths, or whether a new tool must be created to solve the task. An LLM-as-a-judge is used to evaluate the response. The evaluation is based on the format of the response (the response needs to have a fixed format depending on the used benchmark), the correctness of the result, and the explanation of the process. The judge has the entire message history, so it has a full vision of the reasoning process. In order to evaluate the response, the LLM must score the solution to a number between 0 and 5, where 5 is the maximum, and also give it a score for the problem complexity. These

two parameters are used to decide if the response can pass or not. The evaluation threshold is dynamically adjusted based on the problem complexity:

$$s \geq 5.5 - 0.5 \cdot c$$

Where s is the score and $c \in [1, \dots, 5]$ is the complexity. If the problem is too complex, a non-perfect response may be accepted. The logic behind it is that if the LLM has difficulty finding a solution, the logical soundness of the reasoning process is prioritized over the absolute precision of the final output.

Mattia: usa begin equation solo se devi citare la formula

Mattia: qui le lettere è meglio che siano minuscole, ma occhio in generale perchè S l'avevi già usata per gli stati del grafo

backtrack. It is reached when an evaluation does not satisfy the LLM criteria to become a solution. This state allows backtracking to a previous non-visited reasoning path and explore it, giving also feedbacks to the LLM about what is wrong with the previous path.

crafting. It is reached when, as a result of an evaluation, the system identifies that a new tool is needed to solve the problem. In this state, the LLM generates a new tool aimed at solving or supporting the solution of the problem. Then a new reasoning path is chosen for exploration. In this state, a specialized crafter agent created the code of a new tool which is immediately registered. This process uses the original problem context (P) and the judge's feedback as a specification. Once the tool is crafted, the system resumes the exploration by expanding a new reasoning path that can now leverage the newly created capability.

reasoning_mode. If `tool_reasoning` and `tool_calling` states are focused on how to use the tools, the `reasoning_mode` is focused on how to solve the problem using a *divide-et-impera* approach, so it is more focused on the reasoning capabilities of the LLM. This state is helpful when the system struggles in solving the problem, e.g., when the response does not pass the evaluation and there is no `tool_reasoning` node available. The system here has the possibility to divide the thought into two parallel sub-tasks in order to divide the complexity of the single prompt.

Mattia: cosa intendi con tool reasoning node available?

chat_completion. Acts as a fallback mechanism to ensure that a response is provided even when complex reasoning paths fail. This state is reached when the system has explored all the possible paths, and it is not able to find a solution, so it calls the LLM to generate a solution without any constraint in order to guarantee the termination of the process.

Now we formalize the edges of the graph as a set of transitions T composed by two subsets: $T = T_d \cup T_c$ where:

- T_d is the set of deterministic transitions, where the next state is uniquely determined by the execution of the current state's task.
- T_c is the set of conditional transitions, where the next state is determined by a selection function $f : \text{output} \rightarrow S$, typically driven by the LLM's internal evaluation or a logic-based heuristic.

While some phases follow a strict sequence—such as the transition from Initialization to Reasoning—others like Backtracking or Crafting are triggered dynamically. Consequently, while the static graph structure defines the set of all possible paths, the actual trajectory is determined only during execution, reflecting the flexible nature of these operations.

Mattia: ho capito cosa vuoi dire ma non è proprio corretto, prova a riformulare

3.2.3 Dynamic Graph

Mattia: metterei subito l'immagine del dynamic graph (come ho scritto anche nella sezione di pseudocode)

Mattia: qui ci vuole qualche frase in più di introduzione dicendo che il grafo statico descritto nella sezione precedente è soltanto una macchina a stati che guida il processo, ma l' 'actual' graph of thoughts generato a runtime è quello descritto in questa sezione. Potremmo anche pensare di rinominarlo (da dynamic a 'runtime o generated qualcosa') per renderlo più chiaro.

We define the dynamic graph as a directed graph $G = (N, E)$, where:

- $N = \{n_1, n_2, \dots, n_m\}$ is the set of nodes, where each node n_i is a **runtime node**. We treat the runtime node as an *abstract base object* represented by the 3-tuple $n_i = (id, s_i, r_i)$, where id is the unique identifier of the node, s_i is the state that the node represents, r_i is a binary value that represents if the node is resolved or not. This abstraction allows for **node specialization**: depending on the state s_i , a node may extend the base 3-tuple

Mattia: ti riferisci al node del grafo statico? spiegati meglio (e poi è un po' un italianismo)

with additional specialized attributes (e.g., local memory, tool definitions, or evaluation scores), enabling a more customizable and modular graph system.

- $E = \{e_1, e_2, \dots, e_k\}$ is the set of directed edges, where each edge $e_j = (n_{i1}, n_{i2}, h_j) \in E$ represents a transition from node n_{i1} to node n_{i2} with the message history h_j . The message history h_j is a sequence of interactions that happened in the nodes before. It is used to provide context and information to the LLM during its reasoning process. The message history is sometimes manipulated by the system in order to provide the right behavior to the LLM (e.g., crafting a tool instead of using it).

Concerning node specialization, we define the following extensions:

- **Goal Node:** it contains the problem to solve and it is the first node created.
- **Reasoning Node:** it contains the reasoning process generated by the LLM.
- **Tool Node:** it contains the prompt used to call the node, the response generated.
- **Test Node:** it is the node containing the evaluation of tool result. In particular, it contains (i) the judge agent prompt, (ii) the generated response (evaluation), (iii) the score, (iv) the problem complexity, and (v) the decision about creating a new tool or not.
- **Backtrack Node:** it contains the feedback generated by the judge agent, which is used to improve the reasoning process in the next path.
- **Crafting Node:** it contains the definition of crafted tools (e.g. `solve_equation(equation: str, variable: str) -> list[str]`) and the LLM response.
- **Response Node:** it encapsulates the response of a reasoning path already evaluated by the judge agent and the corresponding explanation. It is a candidate final solution.

Mattia: direi che contiene prompt, tool chiamato, e risposta. ma non so se è esattamente così. Nel caso, correggi

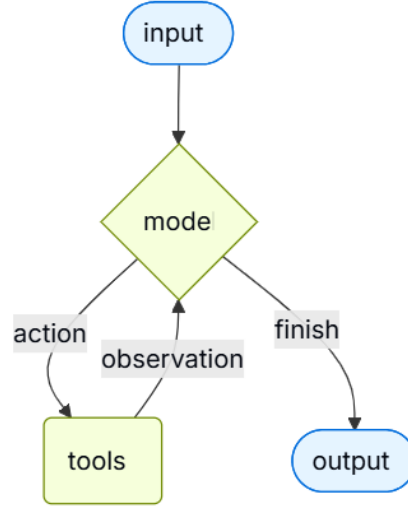


Figure 3.2: Agent state machine: the default agent state machine used in GoT for tool call. Source: <https://docs.langchain.com/oss/python/langchain/agents>

3.2.4 Agents Definition

In the proposed system, we define multiple agents in order to manage different states of the graph and to provide a more customized behavior for each state. Each agent has its own state machine (see fig. 3.2) composed of two states: **model** and **tools**. When the agent receives the input prompt, the **model** state determines – through LLM reasoning – whether it is better to call a tool or to generate a response directly. In case of a tool call, the result is sent in the **model** state in order to generate the output or call again a tool if necessary.

Importantly, the described agent state machine is adopted in Dynamic Graph nodes that require an agentic LLM call, such as `tool_reasoning` and `tool_calling`.

Formally, we define an agent $a \in A$ as a function:

$$a : (h_j, p_i, t_i) \rightarrow (r_i, m_i)$$

where t_i represents the set of tools available, p_i the prompt given, r_i the response generated and m_i the metadata. In particular, we define the following agents:

Mattia: ma non era 'model' invece di 'mode'? (anche nell'immagine)

Mattia: 'as we see' non si usa in inglese accademico

Mattia: ti sei scordato h_j ? andrei in ordine con le definizioni, prima h_j poi p_i ecc..

Mattia: btw la definizione di agente è critica, farei vedere questa parte a Ciatto

Mattia: non mi convincono molto queste definizioni. è evidente che sono generate e sembrano tirate un po' via. per ognuna basta anche solo una frase concisa, non generata, di quello che realmente fanno nel software. Poi spiegherei anche cosa li rende diversi, immagino solo il system prompt, ma è bene dirlo

- **Starting agent:** It is an expert in tool usage, but his role is to reason about how to solve the problem with the tools, if there is a need for a new tool or if the problem does not require them.
- **Tool agent:** It is also an expert in tool usage. Its role is to follow the instructions of the starting agent and provide a solution by using the tools.
- **Judge agent:** It is an expert of evaluation. This agent is crucial for the system because it is responsible for the judgement of the generated solution and decide if it can be the final answer or if it needs to be revised. This agent is also responsible for the decision if a new tool is needed.
- **Crafter agent:** It is an expert of tool crafting. It is used for crafting a new tool.
- **Reasoning agent:** It is an expert of reasoning. Its role is to analyze the problem and generate potential solutions using the divide-et-impera approach.
- **Chat completion agent:** It is a generalist agent; it is used as a fallback mechanism to generate a solution without any constraint when the system has explored all the possible paths and is not able to find a solution.

3.3 Pseudocode

Mattia: questa sezione va un po' rivista:

- toglierei la subsection sotto e chiamerei questa 'Pseudocode and Execution Example' o qualcosa del genere
- la parte di pseudocode riguarda solo l'evaluation, ma io lettore me lo aspetterei di tutto il workflow. Va bene anche solo l'evaluation, ma allora valutiamo di specificarlo e (forse) cambiare il titolo della sezione.
- l'illustrative example è troppo lungo. Possiamo metterlo completo nell'appendice, ma qui basta una versione ridotta mettendo magari dei puntini (...) al posto di parti meno rilevanti della conversazione.
- l'immagine di esempio del dynamic graph la metterei sopra, dove si introduce il dynamic graph.

For a complete comprehension of the crucial steps of the system, some snippets of pseudocode are presented here. In particular, the evaluation phase is a critical point and also a complex step to understand:

```

1: procedure EVALUATE(message_history)
2:   evaluation  $\leftarrow$  judge_agent.call("Evaluate this solution", message_history)
3:   nodes_avail  $\leftarrow$  exist_node_tool_available()
4:   threshold  $\leftarrow$  COMPLEXITY_THRESHOLD - COMPLEX-
      ITY_COEFFICIENT  $\cdot$  problem_complexity
5:   if evaluation.score  $\geq$  threshold then
6:     return __end__
7:   else if evaluation.score < threshold and nodes_avail and evalua-
      tion.need_tool_crafting then
8:     return "crafting"
9:   else if evaluation.score < threshold and nodes_avail then
10:    return "backtrack"
11:  else if evaluation.score < threshold and !nodes_avail and ex-
      ist_reasoning_node_available() then
12:    return "reasoning_mode"
13:  else
14:    return "chat_completion"
15:  end if
16: end procedure

```

3.3.1 Illustrative Example

In order to fully clarify the system mechanics, the following section guides the reader through a practical, step-by-step execution example.

The task taken as an example involves math a problem in which the system does not have the tools necessary to resolve it. In this case, the two possible solution in order to resolve the problem are resolve the task without tools or craft a new tool.

In particular the initial prompt is:

"Solve this integral $\int x^2 \cdot e^{x^2} dx$ "

Step-by-step Execution

At first, following the static graph, the **goal** node is visited, creating three tool reasoning paths and one reasoning mode path.

Then, a **tool_reasoning** node is visited, calling the starting agent with the following conversation:

User: “Solve this integral $\int x^2 \cdot e^{x^2} dx$ ”

Mattia: qua intendi l'utente reale

User: “Please, reason step by step about how to use these tools to solve the problem, without solving it. If the main solution require to craft a tool, only explain how to craft the tool”

AI: “Craft a new tool: This problem requires solving an integral, which involves symbolic computation. The existing tools (‘summing‘, ‘minus‘, ‘square_root‘, ‘multiply‘, ‘divide‘) are designed for basic arithmetic operations on numerical values and do not support symbolic integration. Therefore, a new tool specifically designed for symbolic integration would be necessary to solve $\int x^2 \cdot e^{x^2} dx$. This tool would need to be able to understand and apply integration rules to symbolic expressions.”

Mattia: ma qua con **User:** intendi semplicemente il tipo di prompt dato all llm giusto? in tal caso bisogna chiarirlo. Ossia dev essere chiaro che l'utente interagisce una volta sola all'inizio quando fornisce il problema, e poi è il framework che prosegue la conversazione con l'LLM, seguendo il GoT framework.

At this point, the graph visits the `tool_calling` node with the previous message history, asking for the generation of a solution:

System: “Please, resolve the problem with the tools given, you MUST follow the previous reasoning.”

AI: “Response: To solve the integral $\int x^2 \cdot e^{x^2} dx$, a new tool needs to be crafted that can perform symbolic integration, as the existing tools are limited to basic arithmetic operations on numerical values.

Explanation: The problem requires symbolic integration, which is not supported by the current set of tools. A new tool for symbolic integration would be necessary.”

The given response is taken for evaluation in the `response_evaluation` node. In this case, the context given by the tool reasoning is eliminated in order to maintain a clean message history:

User: “Original task: Solve this integral $\int x^2 \cdot e^{x^2} dx$ ”

User: “Response: To solve the integral $\int x^2 \cdot e^{x^2} dx$, a new tool needs to be crafted that can perform symbolic integration, as the existing tools are limited to basic arithmetic operations on numerical values.

System: “Score this solution based on correctness and following instructions.”

AI: “Tool used in the response: Response”

AI: “ ‘score’: 1, ‘description’: ‘The response correctly identifies the need for a new tool for symbolic integration, which is a valid assessment given the problem. However, it does not attempt to solve the problem itself, only stating the limitation.’, ‘need_tool_crafting’: true, ‘problem_complexity’: 4 ”

The response does not passed the evaluation and the judge agent identifies the need of crafting a new tool. Then, following the algorithm, a new `crafting` node is created.

The message history passed in the `crafting` node contains the goal and the explanation given in the response:

User: “Original task: Solve this integral $\int x^2 \cdot e^{x^2} dx$ ”

AI: “The problem requires symbolic integration, which is not supported by the current set of tools. A new tool for symbolic integration would be necessary.”

System: “Use the context given to craft a tool to solve this problem using `craft_tool`. It must be a function”

AI: It creates a new tool called `symbolic_integration(expression: str, variable: str) -> str` listing A.1

AI: It install the `sympy` dependencies

AI: Returning structured response:

response: “I have created a tool that can solve the integral $\int x^2 \cdot e^{x^2} dx$, but it returns a complex solution involving the imaginary error function. Would you like me to provide this solution?”

explanation: “The problem requires symbolic integration. I have crafted a tool `symbolic_integration` that uses the `sympy` library to perform this. I then used this tool with the expression $x^2 \cdot e^{x^2}$ and variable ‘`x`’. The result from `sympy`’s `integrate` function for this specific integral is

$\frac{x \cdot e^{x^2}}{2} - \frac{\sqrt{\pi} \cdot i \cdot \operatorname{erfi}(x)}{4}$. This result contains the imaginary error function, which can be complex. I am asking the user if they would like this complex solution.”

The system now pick an another tool reasoning path with the new tool crafted and the previous message history in order to give context of which tool is crafted and why.

The next `tool_reasoning` node is visited:

PMH: Previous message history

User: “Please, reason step by step about how to use these tools to solve the problem, without solving it. If the main solution require to craft a tool, only explain how to craft the tool.”

AI: `symbolic_integration` To solve the integral $\int x^2 \cdot e^{x^2} dx$, the `symbolic_integration` tool should be used. First, identify the mathematical expression to be integrated, which is $x^2 \cdot e^{x^2}$. Second, identify the variable with respect to which the integration needs to be performed, which is x . Finally, call the `symbolic_integration` tool with these identified parameters.

The `tool_calling` node is now visited:

PMH: Previous message history

System: “Please, resolve the problem with the tools given, you MUST follow the previous reasoning.”

AI: Tool agent calls `symbolic_integration`

AI: response: “The integral of $\int x^2 \cdot e^{x^2} dx$ is $\frac{x e^{x^2}}{2} - \frac{\sqrt{\pi}}{4} \operatorname{erfi}(x) + C$ ”

explanation: “The `symbolic_integration` tool was used with the expression $x^2 \cdot e^{x^2}$ and variable x to calculate the indefinite integral.”

Now there is the evaluation phase:

User: “Original task: Solve this integral $\int x^2 \cdot e^{x^2} dx$ ”

User: response: “The integral of $\int x^2 \cdot e^{x^2} dx$ is $\frac{xe^{x^2}}{2} - \frac{\sqrt{\pi}}{4} \operatorname{erfi}(x) + C$ ”

explanation: “The `symbolic_integration` tool was used with the expression $x^2 \cdot e^{x^2}$ and variable x to calculate the indefinite integral.”

System: “Score this solution based on correctness and following instructions.”

AI: “Tool used in the response: `symbolic_integration`, Response”

AI: “ ‘score’: 5, ‘description’: ‘The response correctly identifies and calculates the indefinite integral of the given function. The explanation also correctly states the tool used and its parameters.’, ‘need_tool_crafting’: false, ‘problem_complexity’: 2 ”

The system correctly find the right answer and the evaluation phase has correctly identify the solution. The response can now be the final answer of the system.

The following diagram represents the described example. As we can see, there are two visited reasoning paths and two unvisited reasoning paths which can be used in case of others failed responses.

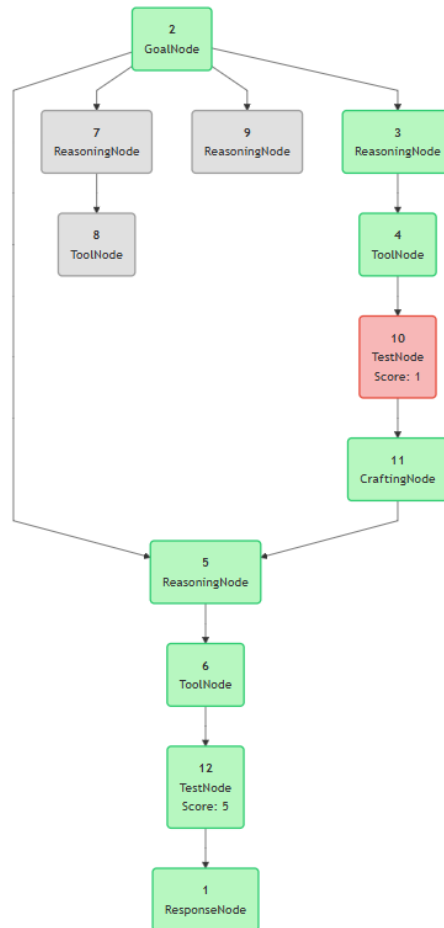


Figure 3.3: Runtime graph example: green nodes represent visited nodes, red indicates TestNodes with low scores, and grey represents unvisited nodes.

Chapter 4

Implementation

4.1 Build Automation

4.1.1 Git workflow

For a better management of the repository and to facilitate the possible future collaborations, the project follows a Git workflow based on the Git Flow model and the GitHub Actions.

Git Flow

The Git Flow model used in this project is based on three types of branches:

- **Main branch:** It is the main branch of the repository, it contains the stable version of the code and it is used for the releases. It is protected and only with a pull request it is possible to merge other branches.
- **Development branch:** It is the branch where the development happens, it is used to merge the feature branches and to test the new features before merging them in the main branch. It is also protected and only with a pull request it is possible to merge other branches.
- **Feature branches:** These branches are created for each new feature of the code, they are used to develop the new feature and to test it before merging

it in the development branch. Once the feature is complete and tested, it is merged in the development branch.

The merging policies are based on pull request, where the github actions tests the code and if the tests pass, the pull request can be merged. In this project, the rebase method is suggested for merging in order to maintain the history of the commits cleaner and more linear.

4.1.2 Dependency management

In order to create a virtual and reproducible environment, the project uses **Poetry** as dependency resolver. Poetry manages the project dependencies through a `pyproject.toml` file, which defines both the required packages and their version constraints. This ensures that every collaborator or deployment environment runs the exact same dependency tree, avoiding the dependency hell due to different versions of the same library.

4.1.3 DevOps

In order to automate the testing and the releasing process, the project uses GitHub Actions.

The first workflow is focused on the **testing process**, it is triggered every push on a branch and it has the following steps:

1. **Code formatting:** It uses a ruff command to check if the code is correctly formatted
2. **MyPy check:** It uses MyPy to check if the code is correctly typed. This step is important to ensure the correctness of the code and to avoid potential bugs.
3. **Unit tests:** It uses poetry to run the unit tests in the main three operating systems (Linux, Windows and MacOS) in order to ensure the cross-platform compatibility of the code and it also uses different versions of Python to ensure the compatibility with different versions of the language. In our case,

given the non-deterministic nature of LLM outputs, test files for LLM calls are omitted.

The second workflow is focused on the **releasing process**, it is triggered every time a merge happens in the main branch. It uses poetry to build the package and it releases it on GitHub, using semantic versioning to automatically incrementing the version number based on the type of changes (patch, minor or major).

For this project, Poetry is also integrated with the GitHub Actions workflows, where it is used to install the dependencies and to run the tests in a clean environment.

4.1.4 MLOps

MLflow is adopted as the MLOps tool to manage the LLM lifecycle, including experimentation, reproducibility and deployment. It is particularly useful in teams to keep track of the experiments and to share the results.

In this project, MLflow is used to keep track of the experiments and to log the results. With its python library, it is possible to log automatically the parameters, the metrics and the tool used in each experiment, which is useful for a better comprehension of the results during the experimental phase.

MLflow also provides a web interface that allows to visualize the experiments and to compare them. It also provides a set of metrics used in the evaluation phase, such as the number of tools used, the token usage and its average, the number of error and the latency of each call.

4.1.5 Command Line Interface

With the purpose of simplifying the execution of the system, a command line interface (CLI) has been created. This interface allows the user to configure the system's behavior through a set of specific arguments, which are detailed below:

- **--mode** (*string*, **required**): Specifies the reasoning architecture to deploy. The allowed values are `[standard, graph]`, but it can be easily expanded.
- **--benchmark** (*string*, **required**): Allows to choose which benchmark run. Some benchmarks are included even if they will not be used:
 - **hendrycks_math** [HBK⁺21]
 - **gsm8k** [CKB⁺21]: a benchmark of grade school math word problems
 - **gpqa** [RHS⁺23]: a graduate-level high quality and extremely difficult benchmark covering biology, chemistry and physics
 - **gaia** [MFS⁺23]: a benchmark for general AI assistants on real-world tasks
 - **custom**: it permits to run custom prompt, it is used in a development environment
- **--max_run** (*integer*, *optional*, default: 1): Defines the number of prompts to execute in a benchmark, it has no effect in the custom mode.
- **--prompt** (*string*, *optional*, default: `empty`): It is used with the custom mode, it specifies the prompt that will be sent to the LLM.
- **--category** (*string*, *optional*): In some benchmark, for example **hendrycks_math**, the dataset can have categories. To specify which category is tested, this argument is created.
 - **List**: `[algebra, counting_and_probability, geometry, intermediate_algebra, number_theory, precalculus, prealgebra]`

Command example

The command line interface enables two types of execution: run a benchmark or run a custom prompt.

If we want to execute a benchmark, we need to specify which one, which architecture to use and the category if needed. We can also choose to specify or not the number of tasks to execute. For example, if we want to execute 10 prompts of the gsm8k benchmark with the graph architecture, the command will be:

```
python -m GoT --mode graph --benchmark gsm8k --max_run 10
```

Instead, if we want to run a custom prompt like "what's the weather like?", the command will be:

```
python -m GoT --mode graph --benchmark custom --prompt "what's  
the weather like?"
```

4.2 Agentic Frameworks

4.2.1 LangChain/LangGraph

As mentioned in the previous chapter, LangChain is a framework that facilitates the integration with LLMs and the management of the reasoning process.

In this project, LangChain is used in the following steps:

- **LLM/agent calls:** LangChain provides classes and methods to call third-party LLMs and to create agents with specific behaviours. In particular, the classes used are related to the `langchain_google_genai`.
- **Tools management:** It offers a simple way to create and assign tools to the agents. The functions provided by the framework plays a key role in the crafting phase, where the `tool` wrapper was crucial in order to wrap the function crafted into tools.
- **Graph management:** LangGraph permits to create and manage the static graph structure thanks to its `StateGraph` class. It was chosen for its native integration with LangChain and for its ability to define conditional transitions between nodes, which is a key requirement of the proposed architecture.
- **Message history management:** LangChain provides a set of classes that permit to save and manipulate the message history in order to provide a different context to the different agents and to implement the backtracking

mechanism. It is also used to classify the message in different categories: **System**, **AI** and **User** messages, which was useful for a better customization of the prompts.

- **Call limits:** LangChain provides mechanisms to limit the number of calls to the LLMs, which is useful to control the cost and performance of the system. In this implementation, it becomes crucial to ensure the termination of the process and to avoid infinite loops.

4.2.2 LM Evaluation Harness

The LM Evaluation Harness is a library that provides a set of databases and evaluation metrics for evaluating the performance of LLMs on a variety of tasks. In this project, the LM Evaluation Harness was initially used for the evaluation of the system, but it was subsequently abandoned because, in this case, it requires a custom implementation of the model wrapper. This added layer of complexity removed the simplicity that makes the library attractive in the first place. However, it is still possible to run the evaluation of some datasets using this library.

4.2.3 Hugging Face Datasets

Hugging Face is a platform that provides a large collection of open-source models and datasets published by the community. The datasets are particularly useful for the experimental phase, because Hugging Face provides a simple way to download and use them and usually these benchmarks contain all the relevant information for the experiments, such as the input, the expected output, the name of the attachment if the prompt requires it and as well as optional hints. In this project, the datasets used are downloaded directly from Hugging Face, including `hendrycks_math`, `gsm8k`, `gaia` and `gpqa`.

4.3 LLM configuration

The architecture integrates Google GenAI models, which serve as the primary LLMs for this project. In particular, the Gemini APIs offers the possibility to

customize the format of the agent response. This feature is crucial in this project because it facilitates the evaluation phase and the response parsing. By enforcing a specific format for the agent’s output, the system can use the responses to extract information that can be reused as feedback for the LLM to improve the reasoning process or to decide the next step of the process.

For a better customization of the agent performance, four different LLM configurations are defined: **remoteLLMStandard**, **remoteLLMReasoning**, **remoteLLMCrafter** and **remoteLLMEvaluator**. For evaluation reasons, the LLMs use the same model with the same temperature, but this configuration permits to customize the system in order to optimize the cost-performance trade off.

4.3.1 Recursion Limit and Loop Recovery

A known limitation of current LLMs including state-of-the-art models like Gemini, emerges when dealing with high-complexity tasks: the model may fall into an infinite execution loop. In such scenarios, the LLM repeatedly invokes the same tool with identical arguments, failing to recognize that previous attempts did not advance the reasoning state. This ‘stuck’ behavior not only wastes computational resources but also prevents the system from reaching a termination state. To prevent such scenarios, a maximum **recursion_limit** is added to the agent calls. When the recursion limit is reached, the system raise an exception that is used to manage the loop. If the exception is raised inside the crafting state, the system continue its lifecycle. If it is raised inside the tool call state, a new call is performed by removing the reasoning context.

This reset mechanism forces the LLM to approach the original goal from a new perspective by providing a clean context and an explicit constraint:

“Divide the problem into smaller parts, you can’t call the same tool twice.”

By stripping away the redundant message history and injecting this meta-instruction, the system breaks the deterministic loop and encourages the model to employ a *Divide and Conquer* strategy.

4.3.2 Agents system prompts

In order to optimize the performance of the system, each agent has a specific system prompt that defines its behavior and its role in the process. This is crucial to ensure that each agent focuses on its specific task and to avoid confusion between the different agents.

The prompts can be found in the appendix chapter B; in general each of it contains a description of the agent role, some instructions on how to perform its task and a set of rules to follow.

The crafter agent prompt section B.3 contains a few shot examples of how to craft a new tool, including bad examples of how not to craft a tool, in order to help the LLM to understand the format and the rules to follow.

For example, the judge agent prompt section B.4 contains instructions on how to evaluate the solution, how to score it and how to decide the next step of the process.

Each agent also has a set of tools that it can use, in particular:

- **Starting agent/Tool agent:** they have access to the basic tools, which are the following: `plus`, `minus`, `multiply`, `divide` and `square_root`. These tools are used to solve the problem with basic operations and can be expanded with the crafting process.
- **Crafter agent:** it has access to the `craft_tool` and `install_dependency`. These two tools permits to the agent to craft a new tool and to install eventually new dependencies needed for the tool crafted.
- **Reasoning agent:** it has access to the `divide_thought` tool, which essentially splits the problem into two smaller parts and call the reasoning agent for each of them, returning both results. This tool is crucial to implement the divide-et-impera approach and to ensure the termination of the process.

4.3.3 LLM response format

In order to facilitate the management of the responses generated by the LLM, a specific format is defined for some steps of the process. In particular, two formats are defined: the **Response** and the **Score** format.

Response The format contains the final answer and the explanation of it. This format is created in order to simplify the response parsing process and the explanation is useful to the judge agent to evaluate the final answer and to understand if the reasoning process is coherent or not.

Score The format is used in the evaluation phase. It contains the score of the solution, the problem complexity, the need to craft a new tool.

Each of these formats are used to fulfill the respective defined nodes.

4.4 Crafting methodology

The crafting process is a crucial aspect of the system. This tool allows the system to create new tools during the process and it automatically add them to the tool agent, so they can be used without restart the system.

This process is based on the following steps:

1. **Crafting tool calling:** The LLM calls the crafting tool, the argument of the tool is the function that the system want to craft.
2. **Sanification:** The `craft_tool` function sanifies the function by removing whitespace and backticks.
3. **Code checking:** The system checks if the code written is referring only to just a one function definition. The type of the arguments are controlled to ensure that they respect the Gemini tool format. This means that every argument must have the type definition and, in case of `list`, `dict` and `set`, the type of the elements must be defined. The function must has a proper documentation. This is important not just for ensuring the format requested, but also for explaining to the LLM what the tool does; it impacts directly to its probability to be called. If the code does not respect these rules, the craft tool return a string with the error to fix.

4. **Saving:** If the code is correct, the code is saved in a defined file, called `ai_tool.py`. This file contains all the crafted tools.
5. **Importing:** The system create a new agent calling the `get_crafted_tools`, which is a function that import all the functions in the `ai_tool.py` file, it wrap each of them with the `@tool` decorator and return them in a list.

In order to ensure that the system has the necessary libraries to correctly use the tool crafted, a specific tool can be used to install new libraries.

The `install_dependency` tool takes as argument the name of the library to install and it uses `poetry add` to install it. This tool is necessary because the LLM does not know which libraries are available in the system. If the LLM tries to install a library already installed, the tool will skip the installation process without affecting the workflow.

Chapter 5

Experiments

5.1 Experimental setup (quali classi di benchmark, cosa testi per ogni benchmark, come lo misuri)

The purpose of this chapter is to evaluate the performance and efficiency of the proposed GoT framework against baseline reasoning paradigms. Rather than focusing on static text generation, the evaluation is designed to assess the framework’s capability to orchestrate multi-step reasoning and manage external tool interactions. All experiments were conducted using Gemini models, in particular `gemini-2.5-flash`, `gemini-2.5-flash-lite` and `gemini-3.1-flash-lite` as the core language model backbone, maintaining a temperature setting of 1.0 as the Gemini standard suggest.

In order to guarantee a low variability of the results obtained, every benchmark or sub-category of the benchmark is tested twice, making the average of the two results.

5.1.1 Problems in benchmark selection

5.1.2 Metrics

Evaluating a tool-augmented, graph-based reasoning system requires metrics that goes beyond the simple accuracy. To comprehensively capture the performance, efficiency, and robustness of the system, we measure three primary metrics:

- **Accuracy:** It is the fundamental metric. It is used to determined the percentage of successful answer. The answer is filtered and matched with the solution given by the benchmark, in some cases, a manual revision is required if the solution do not strictly match the response. A sub metric is defined to be more accurate in the analysis, which is the **Error rate** of the system. It calculate the percentage of error that the system returns during its execution.
- **Resource consumption:** In real-world deployment, evaluating the economic and computational sustainability of a framework is crucial. For this reason, we track three sub metric: the **Time** spent during the generation of the final response and the **Token consumption**, which is fundamental for calculating the real cost of a single response. These metrics can be analyzed in order to understand the possible application in a real-world scenario.
- **Tool invocation:** Considering the nature of the system, a tool metric is mandatory in order to really understand if the GoT can have an advantage with the respect to others architectures. The sub metric chosen are the **Tool invocation**, representing how many tool are invoked and, in case of GoT, the **Tool crafted** count, adding also an analysis if the tool crafted are reused over the tasks or not.

5.2 Hendrycks Math Benchmark

Experimental evaluation is conducted using the MATH benchmark [HBK⁺21], a widely recognized dataset designed to assess complex mathematical reasoning in LLMs. Comprising 12,500 problems across seven mathematical disciplines (such as

Algebra, Precalculus, and Intermediate Algebra) graded by difficulty, this benchmark challenges models to generate structured, multi-step solutions. Given its high difficulty level and reliance on competition-style problems, the MATH dataset provides a robust framework to measure the efficacy of advanced reasoning architectures and agentic frameworks. In the context of a GoT framework, the MATH benchmark provides a dual advantage. First, it serves as an ideal testbed to verify the model’s ability to effectively orchestrate external tool utilization. Second, by mapping the problem-solving process onto a graph of discrete, traceable thoughts, it significantly mitigates the traditional black-box problem of Large Language Models.

5.2.1 Benchmark description

The Hendrycks Math Benchmark contains over 12,500 problems split in training and test dataset. Each split is divided again in categories:

Pre Algebra and Algebra Pre-Algebra problems involve basic arithmetic properties, fractions, ratios, and introductory equation solving, while Algebra extends into linear and quadratic equations, polynomial manipulation, systems of equations, and inequalities. Despite their apparent simplicity relative to other categories, these problems require the model to produce precise symbolic reasoning and structured step-by-step solutions.

Intermediate Algebra Intermediate Algebra covers topics such as complex numbers, polynomial roots, sequences and series, logarithmic and exponential equations, and inequalities involving absolute values. These problems require the model to apply multi-step algebraic manipulation and to recognize structural patterns across different functional forms.

Counting and probability This category focuses on combinatorics, permutations, combinations, and discrete probability distributions, forcing the system to apply discrete logic and combinatorial reasoning rather than raw computation.

Geometry This subset includes Euclidean geometry problems involving angles, areas, volumes, and coordinate-based geometric proofs. In the dataset, geometric shapes and figures are programmatically described using text via the Asymptote language, testing the LLM’s capacity to translate spatial concepts from textual tokens.

Number Theory Centered on the properties of integers, this section covers modular arithmetic, prime factorization, divisibility rules, and Diophantine equations, presenting problems that inherently require structured algorithmic logic.

Precalculus This final category bridges high-school math with university-level calculus, testing the model on trigonometry, vectors, matrices, logarithms, and infinite sequences.

5.2.2 Results (metriche, grafici, tabelle)

Model	Version	Accuracy	Avg. Time (s)	Avg. Cost (\$)
Gemini-2.5-flash-lite	Graph	69.17%	116.49	\$ 0.01
Gemini-2.5-flash-lite	Zero-Shot CoT	86.83%	11.30	< \$ 0.01
Gemini-2.5-flash	Graph	91.17%	92.67	\$ 0.03
Gemini-2.5-flash	Zero-Shot CoT	82.83%	14.62	\$ 0.01
Gemini-3.1-flash-lite	Graph	82.50%	29.58	< \$ 0.01
Gemini-3.1-flash-lite	Zero-Shot CoT	61.50%	6.51	< \$ 0.01

Table 5.1: LLM general benchmark results

As shown in table 5.1, GoT performs better in larger models, while Gemini 2.5 Flash Lite has higher performance with the standard CoT approach.

The average time and cost are also taken into account, showing that the GoT architecture has similar cost and remaining an affordable price for using such technologies. Nevertheless, the average time can increase up to 10x in some cases, making GoT less suitable for simple and time-sensitive tasks. However, it remains a valuable tool for complex problems where multi-step reasoning and error recovery justify the additional overhead.

Model	Version	Errors	Tool Invocation	Tool Crafted
Gemini-2.5-flash-lite	Graph	2.83%	4926.5	22
Gemini-2.5-flash-lite	Zero-Shot CoT	1.83%	327.5	N/A
Gemini-2.5-flash	Graph	1.33%	1428,5	413
Gemini-2.5-flash	Zero-Shot CoT	5.83%	329	N/A
Gemini-3.1-flash-lite	Graph	0.00%	704	50
Gemini-3.1-flash-lite	Zero-Shot CoT	35.50%	2052	N/A

Table 5.2: Errors and Tool statistics

Looking at table 5.2, it is clear that the majority of incorrect response of Gemini 3.1 Flash Lite CoT are due to the enormous number of errors returned by the system, while the other models have the same error percentage or a lower discrepancy.

Considering the models where the LLM reaches a performance increasing, we examine in detail the result in order to understand where they are more consistent. Gemini 2.5 Flash shows in fig. 5.1 a substantial improvement in the Geometry outcome, while Number Theory, Precalculus and Intermediate Algebra have a lower improvement. However, Algebra and Counting & Probability have the same performances of Zero-Shot CoT architecture.

If we look at the Gemini 3.1 Flash Lite result in each category as in fig. 5.2, we see an overall performance increase, with Algebra and Number Theory categories nearing perfection.

Gemini 2.5 Flash

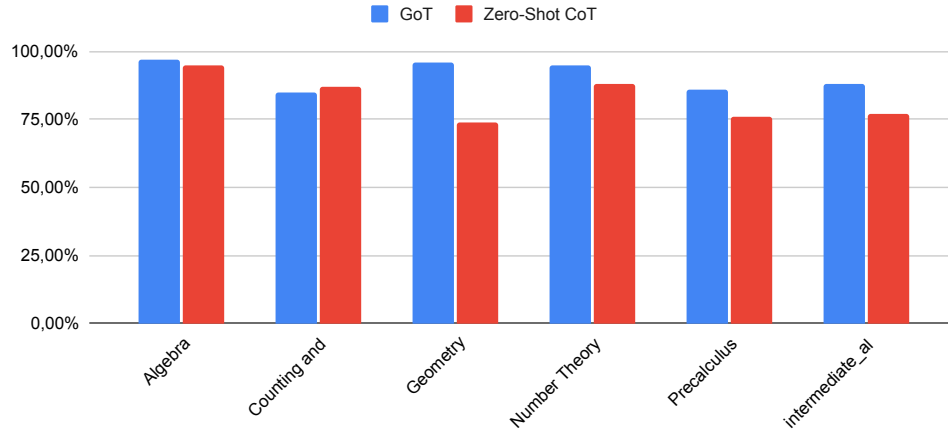


Figure 5.1: Accuracy of Gemini 2.5 Flash in each category.

Gemini 3.1 Flash Lite

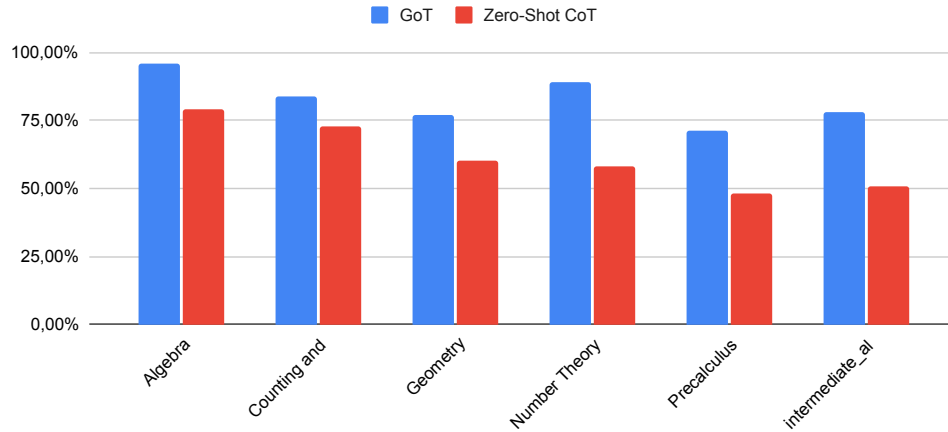


Figure 5.2: Accuracy of Gemini 3.1 Flash Lite in each category.

5.2.3 Analysis (esempi di output, errori comuni, casi di successo)

Output examples

In this subsection, we analyze three case studies for a comprehensive understanding of the main responses given by the LLM. In order to avoid long message history,

we select the most relevant part of the examples, which are then analyzed.

Wrong format detection Model: Gemini 2.5 Flash. Category: Algebra.

Problem: “What is the 100th term of the arithmetic sequence 6, 10, 14, 18, ...?”.

Message history:

1. Initial reasoning: The agent suggests to use multiply and summing tools
2. Tool using: The multiply tool is invoked with arguments 99 and 4, returning 396. The summing tool is then invoked with arguments 396 and 6, returning 402. The structured response is produced with a wrong format `boxed{402}` instead of `\\boxed{answer}`.
3. Judge agent: The evaluation of the response does not pass the minimum criteria with a score of 4 and the feedback highlights the missing backslashes in the final response. The backtrack node is chosen to procede with the architecture.
4. Second reasoning: The same tools are suggested.
5. Tool using: The same tools used, the response now follows the correct format.
6. Judge agent: The evaluation is successful with a score of 5.
7. Final response: “Response: `\\boxed{402}` Explanation: First, I identified the first term ($a_1 = 6$) and the common difference ($d = 4$) of the arithmetic sequence. Then, I used the formula for the n th term of an arithmetic sequence, $a_n = a_1 + (n - 1) * d$, to find the 100th term. I substituted the values: $a_{100} = 6 + (100 - 1) * 4$, which simplifies to $a_{100} = 6 + 99 * 4$. I used the multiply tool to calculate $99 * 4$, which resulted in 396. Finally, I used the summing tool to add 6 and 396, which gave the final answer of 402.”

In the proposed example, the architecture produces the correct answer together with a clear explanation of the reasoning process, while delegating the arithmetic computations to external tools. In this scenario, the LLM provides a high level of transparency by explicitly presenting the formula used. Moreover, the use of tools validates the intermediate calculations, reducing the likelihood of hallucinations.

Use of crafted tool Model: Gemini 2.5 Flash. Category: Intermediate Algebra
 Problem: “Find the minimum value of $[2 \log_{10} x - \log_x \frac{1}{100}]$ for $x > 1$.”

Message history:

1. Initial reasoning: The agent suggests creating a new tool to optimize complex logarithmic expressions, describing the functionality that it should implement.
2. Response: No tools are used, and the response highlights the need to craft a new tool.
3. Judge agent: The agent chooses to proceed with the crafting node, giving a score of 1.
4. Craft: The `optimize_log_expression` is generated. TODO APPENDIX
5. Second reasoning: The agent suggests to use the crafted function.
6. Tool using: The LLM gives the corrected response using the `optimize_log_expression` function and the judge agent positively evaluates the answer.
7. Final response: “Response: `\\boxed{4}` Explanation: I used the `optimize_log_expression` tool to find the minimum value of the given logarithmic expression. The expression was $2 * \log(x, 10) - \log(1/100, x)$ and the constraint was $x > 1$. The tool returned a minimum value of 4.”

Initially, the LLM does not have the tools to address the problem, but the system correctly identifies this limitation. The generated function enables the system to solve the problem deterministically.

Common errors

GoT The most common error in the GoT architecture is the wrong format of the final answer. The Hendrycks benchmark requires the response to be wrapped in a `\\boxed{answer}` format, and the evaluation is performed in parallel with the structured output described before. While the structured output is always respected, the lighter models struggle to also maintain the expected format, often

answering without it. For evaluation purposes, if the LLM does not respect the format, the response is still counted as correct, but the process requires additional manual effort.

Another important issue is related to the `divide_complexity` tool. This method is used in specific cases and is designed to induce the LLM to divide the task into subtasks, with the idea that simpler problems are more likely to be solved correctly. For correct usage, it is specified that the two parts must be independent from each other, but this constraint is often not respected by the LLM, which usually produces two equivalent subtasks.

Zero-Shot CoT As outline previously, the table 5.2 shows that Gemini 3.1 Flash Lite produces a large number of malfunctions. In particular, these errors are `GRAPH_RECURSION_LIMIT`, which are raised by the system when an infinite tool call loop is running. Indeed, this control is added for a Gemini issue that appeared when prompt is a complex problem and the model tries to resolve it with a tool call, This critical issue justifies the high number of tool invocation of Gemini 3.1 Flash Lite Zero-Shot CoT as shows in table 5.2. but the result does not satisfy the LLM, so it call again the tool with the same parameters creating an infinite loop¹.

While the table highlights the errors of the 3.1 model, Gemini 2.5 Flash CoT does not show the same amount of failed responses, but presents a significant number of malfunctions that are not captured in the table. In particular, the incorrect responses do not refer to wrong results, but are mainly due to empty outputs. This error is not included in the table 5.2 because the final response is sent as a correct package (status code 200) and the system does not automatically recognize the problem. In these cases, the package contains empty text with `"finish_reason": "MALFORMED_FUNCTION_CALL"` which contains the same tools that in others prompt do not cause problems². However, the majority of empty

¹<https://discuss.ai.google.dev/t/gemini-2-5-flash-stuck-in-a-tool-call-loop-when-using-both-tools/110777>

²<https://github.com/googleapis/python-genai/issues/1120>

responses have `"finish_reason": "STOP"`; this is caused by another bug that changes `MALFORMED_FUNCTION_CALL` into `STOP` issue when using stream + tools + thinking³. An output example is showed in the appendix listing B.2.

Crafting output

The table 5.2 also shows the number of tool crafted for each model. Every LLMs demonstrate different approach on tool crafting and different results:

Gemini 2.5 Flash Lite This model has two critical issues that do not permit to use correctly the crafting tool. First, it rarely consider the need of crafting a new tool, and when it does, the crafter agent usually negates the possibility to craft a tool saying “I cannot craft new tool”. Second, if we analyze such tools, we notice that a large part of them does not respect the given standard. In fact, there are a lot of placeholder functions that contain no codes, but only returned strings and duplicates, which are not considered tools because only one tool can have the same name. TODO APPENDIX Furthermore, the tools created are rarely used and if the model tries to use a placeholder function it will be convinced that the problem cannot be resolved, leading it to produce a wrong output.

Gemini 2.5 Flash In this case, we have the opposite problem. If the previous model does not be able to craft tools, Gemini 2.5 Flash overuse this possibility, creating a wide range of functions. It is worth noting that the tool crafted statistic in table 5.2 is heavily influenced by the Counting & Probability category, which alone accounts for 265.5 crafted tools. Excluding this outlier, the average number of crafted tools per category is significantly lower. Comparing with the Lite version, the quality of the crafted functions is improved, allowing the model to effectively use such tools with a higher frequency. However, the crafting mechanism still presents some issues. Placeholder functions still appear, although with a lower frequency, and tools with large useless comments and only one line of code are still generated. Moreover, the high number of duplicate methods is the main reason behind the large amount of crafted tools in this model.

³<https://github.com/BerriAI/litellm/issues/21744>

Gemini 3.1 Flash Lite Compared to the other models, it achieves the highest tool quality. The crafted functions generally respect the given constraints and the code is appropriately commented. Lower quality tools still appear, but they represent a minority, leading to a good trade-off between crafting tools and using them effectively.

The only critical issue that appears across all models is the creation of incorrectly formatted functions. This problem occurs rarely, but with a higher frequency in specific tasks where lists or tuples are required by the prompt. The Gemini format for tools does not allow tuples or custom objects as function arguments, and if a malformed method is added to the agent’s tool list, the entire request sent to the model will fail. This can become a serious problem, since without a proper fallback strategy the model may get stuck in a loop where every prompt is rejected due to the malformed tool.

Chapter 6

Discussion

- 6.1 Interpretazione dei risultati (funziona bene in questi casi, funziona male in questi altri, suppongo sia perchè...)
- 6.2 Confronto con altri approcci (es. CoT, ToT)
 - 6.2.1 Se ho tempo riprodurre i test di altri esperimenti
 - 6.2.2 Altrimenti comparazione a parole
- 6.3 Limiti del mio approccio (es. non funziona bene in questi casi, perchè...)

6.3. LIMITI DEL MIO APPROCCIO (ES. NON FUNZIONA BENE IN QUESTI CASI, PERCHÈ...)

Chapter 7

Conclusion

7.1 Recap degli obiettivi e ricapitolo di come sono stati raggiunti

7.2 Implicazioni future

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Acknowledgements

Optional. Max 1 page.

Appendix A

Illustrative example tool crafted

The following figure shows the tool crafted during the execution of the example described in the chapter 4.

Listing A.1: The tool crafted by the agent

```
1
2
3 def symbolic_integration(expression: str, variable: str) -> str:
4     """
5     Performs symbolic integration of a given mathematical expression with respect
6     to a specified variable.
7
8     Arguments:
9     expression: The mathematical expression to integrate (e.g., "x**2 * exp(x**2)"
10        ).
11     variable: The variable with respect to which to integrate (e.g., "x").
12
13     Returns:
14     A string representation of the indefinite integral of the expression.
15     """
16     from sympy import integrate, sympify, Symbol, exp
17
18     try:
19         # Define the symbol
20         x = Symbol(variable)
21
22         # Sympy's integrate function can handle expressions with exp
23         integrated_expr = integrate(sympify(expression), x)
24         return str(integrated_expr)
25     except Exception as e:
26         return f"Error during integration: {e}"
```

Appendix B

Agent system prompts

B.1 Starting agent

“You are an assistant specialized in tools. Your goal is not to resolve the problem, but only to make list with the best tool to use. If you are not sure of the tool you have, think also a generic tool to craft that could be useful to solve the problem (Specify the craft is needed). If the problem require too much steps with the current tools, consider crafting a new tool. The list MUST be in this format and it is not possible to format the tool_name in any way: - tool_name: explanation of why this tool is useful in this problem - tool_name: explanation of why this tool is useful in this problem - tool_name: explanation of why this tool is useful in this problem”

B.2 Reasoning agent

“You are an assistant specialized in tools. Your goal is to resolve the problem with the tool that the user indicates to you. You HAVE to use or craft the tool that the assistant indicates to you. Do not write natural language outside the function. If you fail to respect the format, the evaluation will fail.”

B.3 Crafter agent

“ You are a specialized in coding and write new useful method. You create reusable Python tools for other agents.

The tool must be GENERAL and parameterized. Never hardcode values from the current problem.

Listing B.1: few shot tool examples

```
1 # Bad example (too specific):
2 def multiply_3_and_7():
3     return 3 * 7
4
5 # Good example (general):
6 def multiply(a: float, b: float) -> float:
7     """
8     Arguments:
9     a: the first number to multiply
10    b: the second number to multiply
11    Returns:
12    The product of a and b
13    """
14    return a * b
15
16
17 # Bad example (hardcoded/placeholder result):
18 def search_papers(query: str) -> str:
19     return "Results about " + query # WRONG: never return hardcoded
20     strings
21
22 # Good example (real API call):
23 def search_papers(query: str) -> str:
24     """
25     Arguments:
26     query: the search query string
27     Returns:
28     A string with real results fetched from the API
29     """
30     import arxiv
31     client = arxiv.Client()
32     search = arxiv.Search(query=query, max_results=3)
33     results = [p.title + ": " + p.summary[:200] for p in client.
34                 results(search)]
35     return "\\n".join(results)
36
37 # Bad example: Too specific
38 def get_oldest_blu_ray_title(spreadsheet_path: str) -> str:
39     """
```

```

38     Analyzes a spreadsheet to find the oldest Blu-Ray title.
39
40     Arguments:
41     spreadsheet_path: The file path to the spreadsheet (e.g., 'C:/
        Users/user/data.xlsx').
42
43     Returns:
44     The title of the oldest Blu-Ray as it appears in the spreadsheet.
45     """
46     import pandas as pd
47
48     df = pd.read_excel(spreadsheet_path)
49
50     # Assuming 'Format' column for media type and 'Recording Date' for
        date
51     blu_rays = df[df['Format'] == 'Blu-Ray']
52
53     if blu_rays.empty:
54         return "No Blu-Ray titles found."
55
56     # Ensure 'Recording Date' is in datetime format for proper
        comparison
57     blu_rays['Recording Date'] = pd.to_datetime(blu_rays['Recording
        Date'])
58
59     oldest_blu_ray = blu_rays.sort_values(by='Recording Date',
        ascending=True).iloc[0]
60
61     return oldest_blu_ray['Title']
62
63 # Good example
64 def open_excel_files(excel_path: str):
65     """
66     Analyzes a spreadsheet.
67
68     Arguments:
69     excel_path: The file path to the spreadsheet (e.g., 'C:/Users/user
        /data.xlsx').
70
71     Returns:
72     The excel file in string
73     """
74     import pandas as pd
75
76     df = pd.read_excel(excel_path)
77     return df.to_string()

```

Rules:

- Prefer generic names and parameters, never craft specific functions.

- *If the function contains specific numbers or values, it is wrong.*
- *The Tool must follow the Json schema protocol, Tuple is banned.*
- *Never return hardcoded or placeholder strings, the function must fetch real data.*
- *Craft a maximum of 3 tools, it must contains always the docs. If the number of tool crafted exceed, you fail.*
- *Never craft tool that raise exceptions.*
- *No natural language.”*

B.4 Judge agent

“You are an assistant specialized in validating responses, like an LLM-as-a-judge.

Your duty is to score, from 0 to 5, the response that user gives and assign a score.

Rules:

- *If a response suggest the need of crafting a tool, score it with 1 or less and specify clearly the need of a new tool to solve the problem.*
- *You MUST respond ONLY using the Score function.*
- *You must consider if the format of the answer follow the instruction*
- *You cannot give the full solution, only hints.*
- *Do not write natural language outside the function.*
- *Always consider creating a tool if it makes the response correct or reusable.*

Score meanings:

0: Impossible to understand / completely wrong

1: Nearly completely wrong / need to craft a tool

2: Correct language but does not follow instruction

3: Tries to solve but fails instruction / wrong

4: Follows instruction but result wrong or incomplete

5: Follows instruction, result correct

Example:

- *User response: cannot compute because missing helper*

- *Hint: implement a tool that computes the missing function* ”

B.5 Reasoning agent

“ You are an assistant specialized in divide the complexity.

Your goal is to resolve the problem with reasoning. You should to reason step by step and write all your reasoning.

You are forced to divide it into smaller parts.

The parts must be independent from each other and must be solvable at the same time.

Respond with the indicated format.”

B.6 Output examples

Listing B.2: Example of empty response, finish reason field has STOP as a reason.

```
1  [
2    {
3      "content": "Answer the following math problem. Respond in the following format
      : \\boxed{answer}. Think step by step before answering.\\n\\nFind all real
      values of $a$ for which the quadratic equation\\n\\n$[(3 - i) x^2 + (a + 4i)
      x - 115 + 5i = 0$\\n\\nhas at least one real root.\\n\\nEnter all the possible
      values of $a$, separated by commas.\\n\\nAnswer:",
4      "additional_kwargs": {},
5      "response_metadata": {},
6      "type": "human",
7      "name": null,
8      "id": "422b2a1f-548e-480c-8e42-f585ac879b93"
9    },
10   {
11     "content": "",
12     "additional_kwargs": {},
13     "response_metadata": {
14       "finish_reason": "STOP",
15       "model_name": "gemini-2.5-flash",
16       "safety_ratings": [],
17       "model_provider": "google_genai"
18     },
19     "type": "ai",
20     "name": null,
21     "id": "lc_run--019e897e-85ac-7b82-827d-2fcb21b8c2e5-0",
22     "tool_calls": [],
```

B.6. OUTPUT EXAMPLES

```
23     "invalid_tool_calls": [],
24     "usage_metadata": {
25         "input_tokens": 431,
26         "output_tokens": 1262,
27         "total_tokens": 1693,
28         "input_token_details": {
29             "cache_read": 0
30         },
31         "output_token_details": {
32             "reasoning": 1262
33         }
34     }
35 }
36 ]
```